

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 429 — Past Questions & Materials
by Renance

Course: CHM 429
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 400

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Paper 1 of 3 — CHM 429 - Industrial Chemical Process

Level 400 • Second Semester • PastQuestion

FFURUN, DELTA STATE

RSITY OF PETROLEUM RESOURCES, EFF ,

Riot ge ak COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

IN OCTOBER 2025

SECOND SEMESTER EXAMINATION, 2024/2025 SESSIO ;

COURSE TITLE: INDUSTRIAL CHEMICAL PROCESSES COURSE CODE: CHM429

COURSE UNIT: 2

ANSWER FOUR QUESTIONS ONLY

QUESTION ONE

i. With reference to chalcopryrite (CuFeS_2), explain why some ores have no direct use. as minerals (4-marks}—————

li, Outline the general Processing steps-involved in extracting copper metal from chalcopryrite ore (6 marks)

w)) Explain briefly the term comminution and describe the major equipment used in this process (8 marks)

(c) Using suitable examples, distinguish between industrial minerals and metalliferous ores (6 marks)

QUESTION TWO

(a)Describe the Hall—Hérout process for aluminum production. (10 marks)

(b) Define the following terms (3 marks):

(i) Ore

(ii) Gangue

(iii) Run-of-mine material

(c) i. List three types of run-of-mine mate!

ii, Discuss three advantages and one di

methods in industry.(S5 marks)

U/ QUESTION THREE

(a) Explain the term, Pesticide. (6marks)

(b) Describe any four classes of pesticide and their uses. (12m arks)

(c)_ Write the chemical

tials and give one example each (6 marks)

sadvantage of electrochemical methods over conventional chemical

Fe rete ter

Aicarstructure of DDT and name it. (6marks)

QUESTION FOUR

(a) Before the advent of synthetic dyes, n

dye from a natural source. (10marks)

(b) Outline the process of

local seed. (10marks)

atural dyes were very much in use: describe the process of extracting a

production of a natural flavouring agent from melon or pumpkin seed or any other

(c) State two advantages and disadvantages of the processes and products in (a) and (b) (4marks).

QUESTION FIVE

J a (i) What are inorganic fertilizers? (3 Marks)

(Gl) Write brief notes any two key characteristics of inorganic fertilizers. (10 Marks

Ginn any four (4) examples of inorganic fertilizer (4 Marks)

Aaa

(b). How does inorganic-fertilizer destroy soil? (3 Marks)

(c) Name two advantages and two disadvantage of inorganic fertilizers. (4 Marks)

QUESTION SIX

(a) What is the Process by which cement hardens called? (3 Marks)

(b) Name four (4) stages in the manufacture of cement. (4 Marks)

(c) Discuss any two of the stages in the manufacture of cement. (17 Marks)

END OF PAPER

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Paper 2 of 3 — CHM 429 PQ 2024

Level 400 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFERUN, DELTA STATE
COLLEGE OF SCIENCE
DEPARTMENT OF CHEMISTRY
SECOND SEMESTER EXAMINATION 2023/2024 SESSION SEPTEMBER 2024
COURSE TITLE: INDUSTRIAL CHEMICAL PROCESSES COURSE CODE: CHM 429
INSTRUCTION: ANSWER THREE QUESTIONS ONLY TIME: 2 HOURS

QUESTION ONE

- (a) Briefly explain the following (i) Kiln (ii) calcination (iii) carbonation (iv) hydration. 8marks
(b) i. What are fertilizers? ii. Differentiate between powdered and granulated fertilizer. 9marks.
(c) What are the characteristics a compound must have to be used as a fertilizer? 6marks

QUESTION TWO

- (a) Explain the steps involved in the contact process for the preparation of sulphuric acid. 6marks.
(b) i. State the three methods used in the manufacture of nitric acid. 3marks ii. Discuss one of the methods stated in b (i). 6marks.
(c) Explain the four processes employed in the manufacture of hydrochloric acid. 8marks.

QUESTION THREE

- (a) With relevant chemical equations explain the preparation of paraquat herbicide. 10marks
(b) Outline the steps involved in the preparation of a named food flavouring agent in your named locality. 8marks
(c) Explain the conditions necessary for explosion to occur. 5marks

QUESTION FOUR

- (a) What is a colourant? 5marks
(b) Differentiate between a dye and a pigment. 8marks
(c) With appropriate chemical equations, explain the steps involved in the preparation of azo dye. 10marks.

END OF PAPER

Paper 3 of 3 — CHM 429

Level 400 • Second Semester • PastQuestion

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